

Assignment 3 - Homework & Deliverables

PROJECT 1 | Team Members: Ben B, Gavin O, Jayden O, Sam A

Repository Link: [Link](#)

Coordinate with your team using your new GitHub repository to complete this document before the next class.

Grading Rubric (14 Points Total)

Tech Execution (6 Points):

- **2 pts:** GitHub Repository successfully created, synced, and used by the team.
- **2 pts:** Whiteboard photo properly embedded using Markdown syntax.
- **2 pts:** Mermaid.js code successfully generated, properly formatted in a code block, and renders a valid flowchart.

Mind Map & Concepts (8 Points):

- **4 pts:** Mind Map Breadth (Photo/Code shows robust brainstorming, integrating the blind stimulus with the 25kg challenge, minimum 15+ nodes).
- **4 pts:** Concept Evaluations (Each team member completed their pitch thoroughly, directly addressing how geometry manages the load, and accurately completed the constraint checks).

Section 1: Tech Setup & AI Digitization (6 Points)

1a. GitHub Student Pack (Do this immediately)

- Register for a free [GitHub Student Developer Pack](#). This gives you free access to GitHub Copilot. *(Approval can take time; do this today).*

1b. Whiteboard Photo

(Embed your whiteboard photo here in markdown format: `![Whiteboard Map](link)`)

1c. AI-Generated Mermaid Code

- **Action:** Open **GitHub Copilot Chat** directly inside VS Code. Click the paperclip icon (or drag-and-drop) to upload the photo of your whiteboard.
- **Prompt:** "Act as an expert software engineer. Convert this hand-drawn mind map into valid Mermaid.js flowchart code. Use a top-down (TD) orientation. Ensure the concepts marked with blue are noted."
- **Fallback:** If Copilot struggles to read the handwriting, manually construct your map using

the visual editor at [MermaidChart.com](https://mermaidchart.com), then click "View Code" to copy it.

- **Result:** Paste the Mermaid code below and preview it in VSCode using your Markdown/Mermaid extension.

[Paste your code here]

Section 2: Rapid Individual Concept Evaluations (8 Points)

Each team member claims ONE of the starred concepts from the whiteboard to evaluate against the Phase 0/1 constraints. Ensure your pitches explain the mechanics of the geometric joints.

Concept 1 Evaluator: Gavin O

- **Concept Name:** Stretcher
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load?
(Use Copilot to refine your phrasing)
 - Answer: The Stretcher concept resembles a stretcher, in that it manages the weight load through 3 cardboard lengths running the whole length, like a stretcher. Cardboard connectors run perpendicular to it to provide stability, and will be carried by two people.
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [Y / N]
 - Operable by BOTH 1 AND 2 people? [Y / N]
 - What specific fastener-free geometry handles the connection? (e.g., interlocking tabs, friction-fit, precise folds): **Notched with tabs**
 - What is the biggest risk for failure during the dynamic carry? (creasing, delamination, joint blowout): ____ The Biggest risk is the cardboard handles giving out or the runners folding over _____

Concept 2 Evaluator: Jayden O

- **Concept Name:** Buildability / Construction
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load?
 - Answer: The buildability concept focuses on creating a cardboard structure with no fasteners that can be assembled more efficiently by being used by 1-2 people. It can manage the 25kg load by using interlocking folds and joints to distribute stress throughout the build.
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [Y / N]
 - Operable by BOTH 1 AND 2 people? [Y / N]
 - What specific fastener-free geometry handles the connection?: _We can use interlocking folds to connect the actual handles to the carrier itself or will shape out the handle within the build_
 - What is the biggest risk for failure during the dynamic carry?: _The biggest risk is the quality of how we make the joints and folds especially with the weight already added._

Concept 3 Evaluator: Ben

- **Concept Name:** Foldability
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load?
 - *Answer: A foldable design that can be easily built from already pre measured and cut parts. This allows the design to be transported with ease and built with ease allowing everybody to use the product.*
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [**Y** / **N**]
 - Operable by BOTH 1 AND 2 people? [**Y** / **N**]
 - What specific fastener-free geometry handles the connection?:
 - We will have the cardboard fold in a way that keeps it together, multiple layers of cardboard will likely be required
 - What is the biggest risk for failure during the dynamic carry?:
 - The biggest risk is that the design will fail upon the heavy workload due to either loose connection. Pieces will have to be cut out very precisely for the design to operate effectively

Concept 4 (Evaluator: _Sam Albers_)

- **Concept Name:** __Durability and longevity__
- **The Pitch (1-2 sentences):** What is it and how does it manage the 25kg shifting load?
 - *Answer: Our cardboard carrier uses a stretcher-like design with tall support columns and a double-supported base to distribute weight evenly and prevent bending or shifting. Being so compact the grain should not shit much or at all.*
- **Constraint Checklist:**
 - Cardboard ONLY (no glue/tape/metal)? [**Y** / **N**]
 - Operable by BOTH 1 AND 2 people? [**Y** / **N**]
 - What specific fastener-free geometry handles the connection?:
Folding the cardboard, cutting and sliding in joints, and sliding cardboard through holes to keep it sturdy
 - What is the biggest risk for failure during the dynamic carry?:
The cardboard giving out and not lasting several carries because it becomes weak.

Note: We will use these individual concept evaluations in our upcoming assignment to develop formal design criteria and perform a structured down-selection (using a Morphological Chart or Pugh Matrix). Ensure your GitHub commits are pushed and synced before the next class!

Here is a Mermaid.js conversion of the hand-drawn mind map in a top-down flowchart, with the blue-marked concepts styled in blue:

flowchart TD

```
A["What to build?"]
A --> B["Box"]
A --> C["Users"]
A --> D["Constraints"]
A --> E["Design requirements"]
```

```
B --> B1["Handles"]
B --> B2["Shape"]
B --> B3["Structure"]
B --> B4["Negative space"]
B --> B5["Carry capacity"]
B --> B6["Storage"]
```

```
C --> C1["Red Cross"]
C --> C2["Wheelchairs"]
C --> C3["Compatibility"]
C --> C4["Accessibility"]
```

```
C3 --> C5["Foldable?"]
C3 --> C6["Constructible?"]
```

```
D --> D1["Weight"]
D --> D2["Distance"]
D --> D3["CG / balance"]
D --> D4["Buildable"]
D --> D5["Folded?"]
D --> D6["Compatibility"]
```

```
E --> E1["Form factor"]
E --> E2["Safety"]
E --> E3["Handle / grip"]
E --> E4["Clearance"]
```

```
class B,C1,C2,C3 blueNode
```

```
classDef blueNode fill:#dfeeff,stroke:#1a73e8,stroke-width:2px,color:#0b57d0
```

If you want, I can also produce a more literal version that preserves the sketch's exact grouping and handwritten labels more closely.